Multicollinearity and Outliers

POLS 602
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Review

1. What do robust standard errors change, and what do they leave alone?
2. Why do clustered errors usually make the usual standard errors too small?
3. When would you use WLS instead of OLS with robust standard errors?

Today

- What breaks when predictors overlap?
- What breaks when a few observations dominate the fit?
- How do we detect each problem?
- What should we do about each one?

Multicollinearity

The Problem

- Perfect collinearity: a predictor is an exact linear combination of the others
- Multicollinearity: a predictor is *nearly* a linear combination of the others
- OLS can still be computed
- *Can the data tell the effect of one predictor apart from another?*

Why It Matters

- Standard errors on the affected coefficients are large
- Estimates are unstable: small changes in the data change them a lot
- Individual t-tests can be insignificant while the group is jointly significant
- Coefficients are still unbiased, and predictions are still good

The Variance Inflation Factor

$$\mathrm{VIF}_j = \frac{1}{1 - R_j^2}$$

Where R_j^2 is the R^2 value obtained by regression variable j on all other explanatory variables.

R_j^2	VIF	Standard error multiplier
0	1	1.0
0.50	2	1.4
0.80	5	2.2
0.90	10	3.2
0.99	100	10.0

What Causes It

- Several measures of the same concept
- Variables that are related by construction: totals and their parts, x and x^2, interactions
- Variables that share a trend over time
- Little variation in the sample

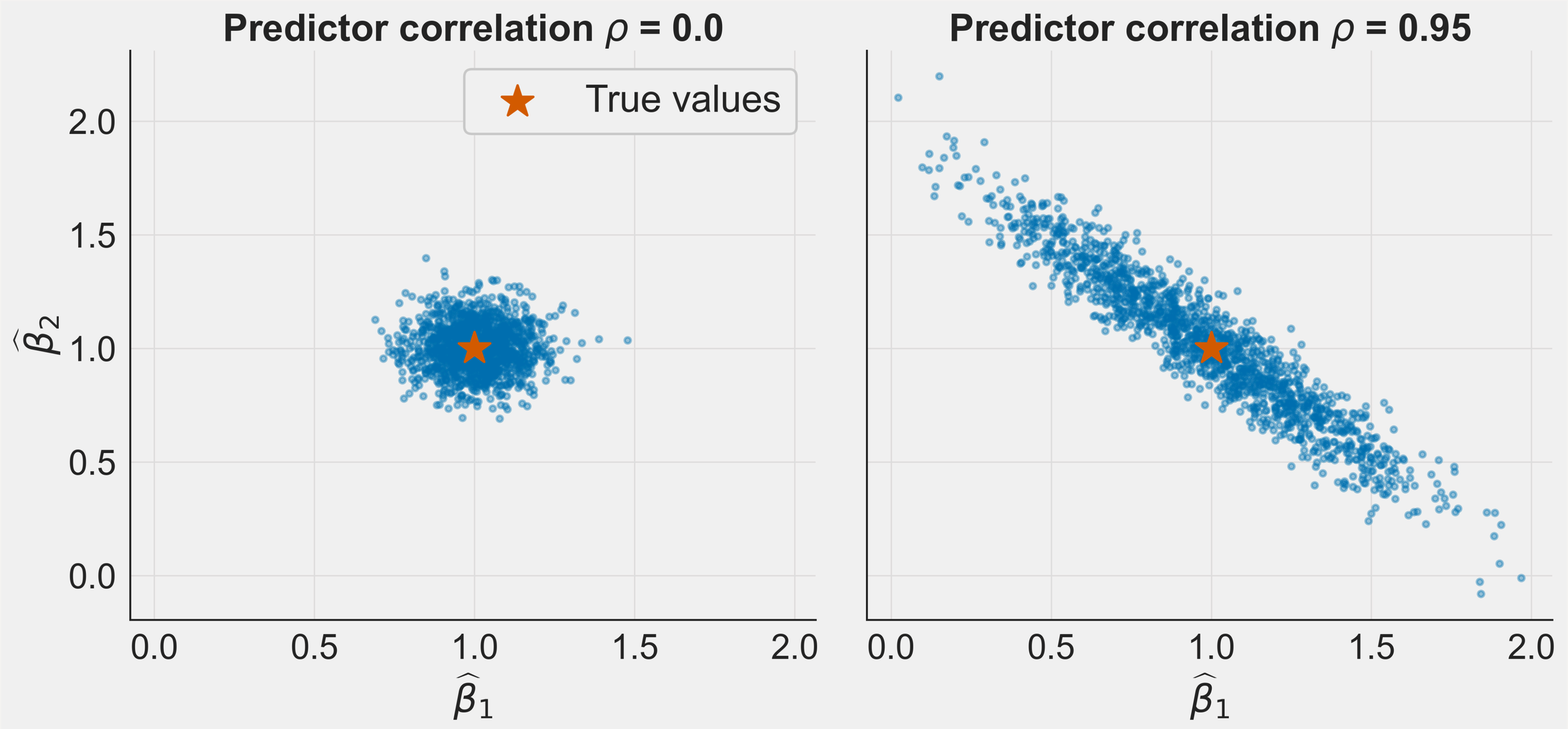
Some intuition on why this matters...

- The data cannot separate the effects of the correlated predictors. But why? What does this mean?
- Think of multicolinearity as a sample size problem.
- Under multicolinearity, “holding the other variable constant” asks about cases that barely occur.
  - E.g. to distinguish the effect of income from wealth, we need to observe cases where income is high, but wealth is low (and the inverse).
  - This rarely occurs, and thus our estimates of the conditional means for such points are noisy and uncertain.

Joint Test Versus Individual Tests

- Individual t-tests ask whether each coefficient is zero
- The F-test asks whether a group of coefficients is jointly zero
- Collinear predictors can fail every t-test and pass the F-test

The Estimates Trade Off



What to Do

- Nothing.
  - If the collinear variables are controls, don’t worry about it.
- Collect more data or better data
- Combine redundant variables. See: PCA, Factor Analysis, Orthogonalized/residualized variables
- Use an F-test
- Generally, you should not drop controls simply because they are collinear.
  - If you do, accept some bias for lower variance.
  - Report results from multiple models.

Outliers and Leverage

The Problem

- **Outlier**: an observation with an unusual y given its x
- **Leverage**: an observation with an unusual x
- **Influence**: an observation that changes the estimates when removed
- *Would the answer change if one observation were missing?*

Why It Matters

- OLS minimizes squared residuals, so large residuals count heavily
- A few points can drive the coefficients
- Conclusions may not replicate on a different sample
- Standard errors and fit statistics can also be distorted

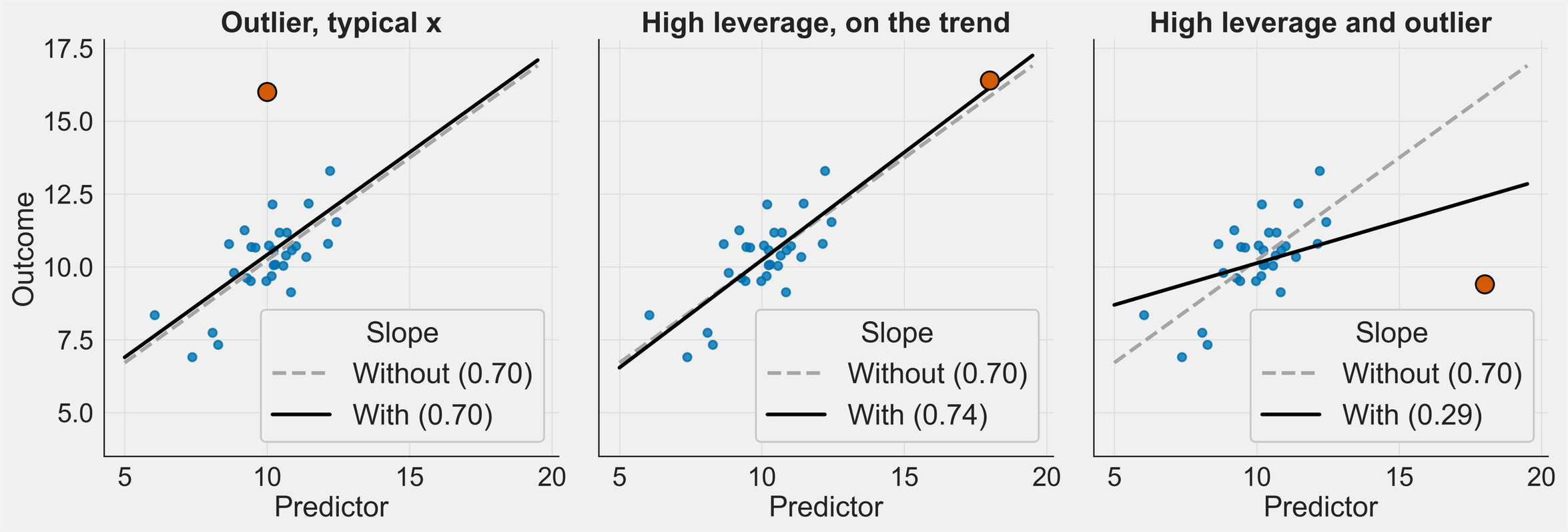
What Causes It

- Data entry or coding errors, such as -999 for missing
- Measurement problems
- Units that are genuinely unusual
- Heavy-tailed outcomes
- A different process, or a model that misses something

What It Implies

- An outlier is information about the model or the data
- It is not automatically an error
- Understand why it is unusual before deciding what to do

Outlier, Leverage, Influence



Leverage: How Much a Point Pulls on Its Own Fit

- Each fitted value is a weighted sum of all n outcomes
- Observation i's own outcome gets the weight h_{ii}
- **Leverage**: h_{ii}, how much y_i pulls on its own fitted value
- Low leverage: the fit at i is set by the other observations
- High leverage: the line is forced to pass near y_i

Where Leverage Comes From: The Hat Matrix

$$\widehat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$$

$$\widehat{\mathbf{Y}} = \mathbf{X}\widehat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$$

$$\widehat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}, \qquad \mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$$

$$\hat{y}_i = h_{ii}\, y_i + \sum_{j \neq i} h_{ij}\, y_j$$

- $\mathbf{H}$ turns outcomes into fitted values: it puts the hat on $\mathbf{Y}$
- Row i of $\mathbf{H}$ holds the weights behind $\hat{y}_i$

Leave-One-Out Influence

- Refit the model without observation i
- Compare with the full-sample fit
- Large changes mean the observation is influential

DFBETAS and DFFITS

- **DFBETA**: $\hat{\beta}_j - \hat{\beta}_{j(i)}$, the change in coefficient j when observation i is dropped
- **DFBETAS**: DFBETA divided by its standard error; flag $|\text{DFBETAS}| > 2/\sqrt{n}$
- **DFFITS**: the standardized change in the fitted value $\hat{y}_i$; flag $|\text{DFFITS}| > 2\sqrt{(k+1)/n}$

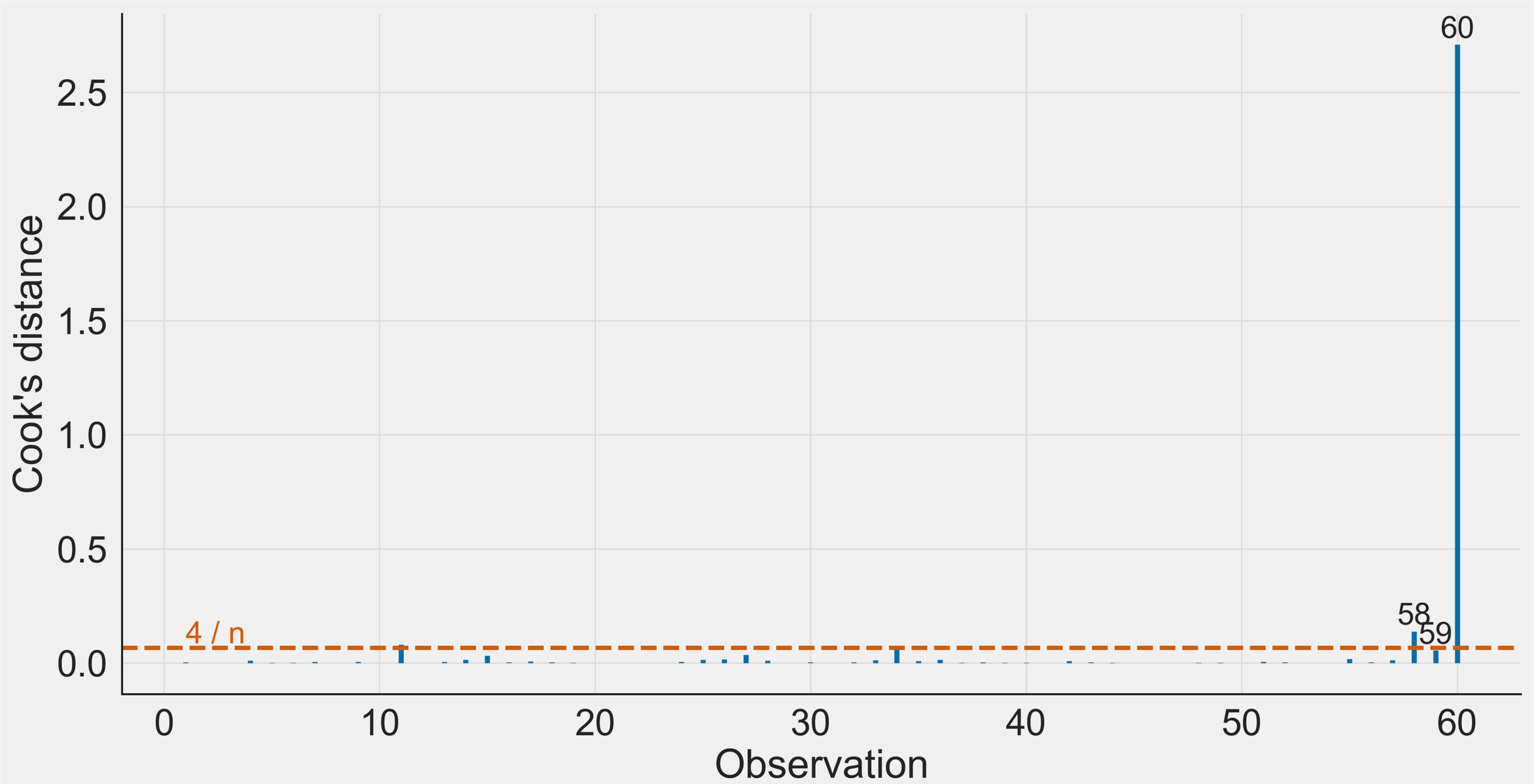
Cook's Distance

$$D_i = \frac{\left(\widehat{\boldsymbol{\beta}} - \widehat{\boldsymbol{\beta}}_{(i)}\right)^\top \mathbf{X}^\top \mathbf{X} \left(\widehat{\boldsymbol{\beta}} - \widehat{\boldsymbol{\beta}}_{(i)}\right)}{(k+1)\,\widehat{\sigma}^2}$$

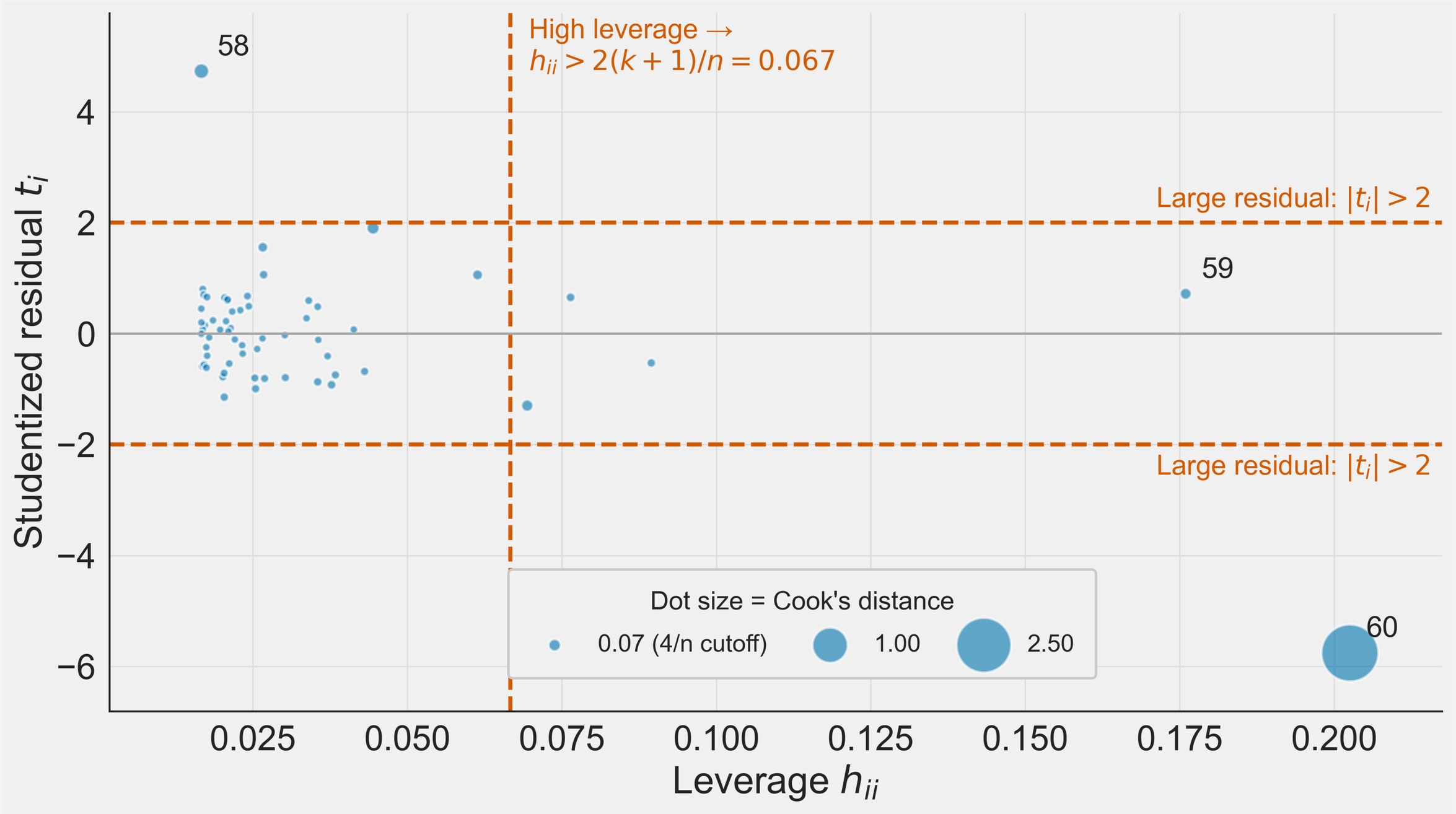
$$D_i = \frac{r_i^2}{k+1} \cdot \frac{h_{ii}}{1 - h_{ii}}$$

- r_i: the internally studentized residual
- Flag $D_i > 4/n$, or a stricter cutoff such as 1

Cook’s Distance by Observation



Leverage vs. Residuals



What to Do

1. Check the data for errors and coding problems
2. Understand why the point is unusual

- Is it part of the same DGP as the rest of the data?

3. Report results with and without the influential observations
4. Reconsider the model’s functional form

Putting It Together

Diagnostics at a Glance

Problem	Tools	Responses
Non-normal residuals	Histogram, QQ plot	Transform, fix the model
Heteroscedasticity	Residual plot, Breusch-Pagan	Robust SEs, WLS
Clustering	Study design	Clustered SEs
Multicollinearity	Correlations, VIF	More data, F-test, variable curation
Outliers and leverage	Cook’s D, DFBETAS	Check, report with and without

Summary

- Multicollinearity inflates variance by the VIF without causing bias
- With collinear predictors, test them jointly
- High leverage is a potential for influence; influence also needs a large residual
- A few influential points can drive the estimates, so check Cook's distance and DFBETAS
- Each problem can be a sign of a wrong model, so ask why before applying a fix

Key Definitions

Key Definitions

- **Multicollinearity**: predictors that are nearly linear combinations of one another
- R_j^2: the R^2 from regressing x_j on the other predictors
- **Variance inflation factor**: $\mathrm{VIF}_j = 1/(1 - R_j^2)$; the factor by which collinearity inflates $\mathrm{Var}(\widehat{\beta}_j)$
- **Ridge regression**: a biased estimator that shrinks coefficients to reduce variance

Key Definitions

- **Outlier**: an observation with a large residual
- **Leverage**: h_{ii}, the diagonal of the hat matrix; how far $\mathbf{x}_i$ is from the center of the predictors
- **Studentized residual**: a residual divided by its estimated standard deviation, adjusting for leverage
- **Influence**: how much the estimates change when an observation is removed

Key Definitions

- **DFBETAS**: the standardized change in a coefficient when observation i is dropped
- **DFFITS**: the standardized change in the fitted value at observation i when i is dropped
- **Cook's distance**: a summary of the change in all coefficients when observation i is dropped; the product of squared residual and leverage terms